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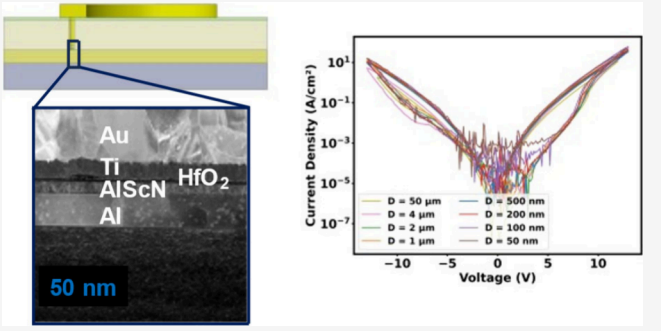
Demonstration of Highly Scaled AlScN Ferroelectric Diode Memory with a Storage Density of >100 Mbit/mm^2

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ABSTRACT: Wurtzite nitride ferroelectric materials have emerged as promising candidates for next-generation memory applications, due to their exceptional polarization properties and compatibility with conventional semiconductor processing techniques. Here, we demonstrate the first successful areal scaling of aluminum scandium nitride (AlScN) ferroelectric diode (FeDiode) memory down to device diameter of 40 nm while maintaining an ON/OFF ratio of >60. Using a 20-nm-thick $Al_{0.64}Sc_{0.36}N$ ferroelectric layer, we evaluate both metal−insulator-ferroelectric-metal (MIFM) and metal-ferroelectric-metal (MFM) architectures for scaled resistive memory devices. Our scaled devices exhibit an enhanced breakdown-to-coercive field ratio exceeding 2.6 due to increased breakdown field. The MIFM devices demonstrate stable 3-bit nonvolatile multistate behavior with clearly distinguishable resistance states and retention exceeding 4×10^4 s at 85 °C. By achieving more than a million-fold areal scaling with enhanced performance metrics, this work establishes AlScN-based FeDiode memory as a highly promising platform for nonvolatile storage with potential for direct integration into CMOS technology.



KEYWORDS: *ferroelectric, storage density, aluminum scandium nitride, memory, BEOL*

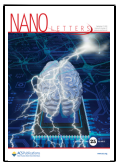
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The development of high-density, nonvolatile memory (NVM) compatible with existing semiconductor logic technology is essential for advancing computing systems beyond current limitations.[1] While volatile memory such as SRAM and DRAM, as well as traditional NVM technologies such as NOR flash memory face significant scaling challenges,[2] emerging ferroelectric materials provide a promising path forward due to their intrinsic nonvolatility, fast switching capabilities, and monolithic integration potential. Aluminum scandium nitride (AlScN) stands out as a particularly promising nitride ferroelectric material for memory applications. AlScN exhibits several critical advantages over other ferroelectric candidates, including a remarkably high remanent polarization exceeding 100 μC/cm^2, substantially higher than oxide-based ferroelectrics such as HfO_2.[3,4] For practical implementation, AlScN thin films can be deposited at temperatures below 350 °C, ensuring compatibility with back-end-of-line (BEOL) CMOS processes and enabling seamless integration with existing semiconductor manufacturing.[5]

The ferroelectric diode (FeDiode) device architecture leverages the inherent ferroelectric polarization-dependent leakage current properties of AlScN to create memory devices with built-in rectification, eliminating the need for additional selector elements typically required in high-density memory arrays.[5,6] However, a critical question remains unaddressed in the literature: Can wurtzite FeDiodes maintain consistent properties such as current density, ON/OFF ratio, and retention when scaled to dimensions necessary for competitive areal memory densities? While theoretical calculations suggest favorable scaling potential,[7] experimental demonstration of functional, scaled AlScN FeDiodes has been lacking.

In this work, we present the first demonstration of area-scalable $Al_{0.64}Sc_{0.36}N$ (hereafter referred to as AlScN)-based FeDiode memory with devices successfully fabricated and characterized from diameters of 50 μm down to diameters of 40 nm (more than a million-fold scaling in the device area). We systematically investigate two device architectures: a metal−insulator-ferroelectric-metal (MIFM) structure incorporating a 4 nm HfO_2 insulating layer and a metal-

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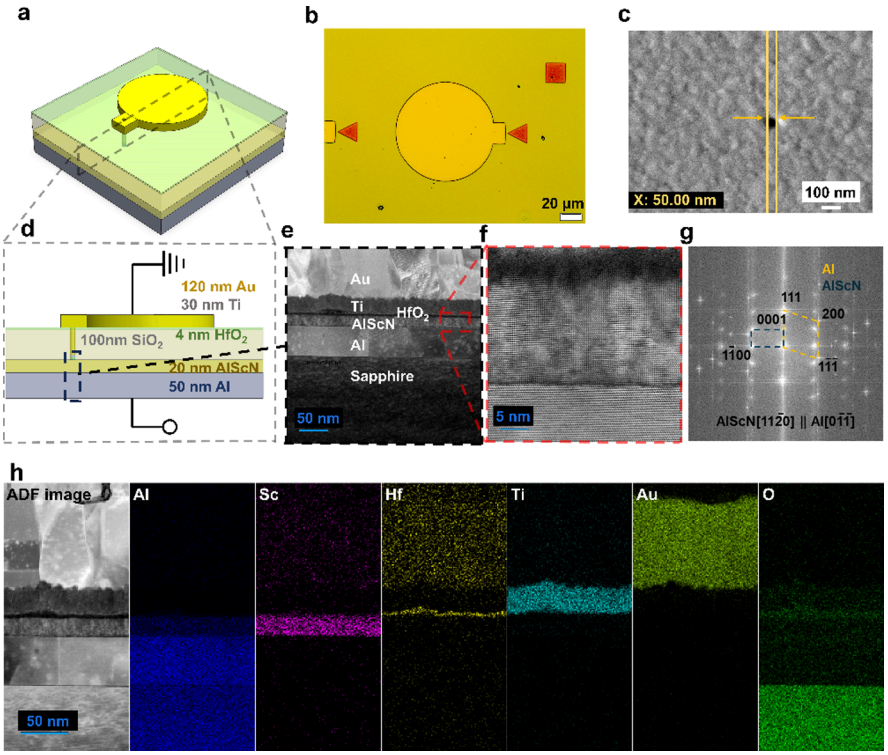


Figure 1. Structural and morphological characterization of the fabricated MIFM device. (a) Schematic illustration of the device layout. (b) Optical image of a fabricated device with a diameter of 4 μm. (c) SEM image showing the surface morphology and the structural details of a 50-nm-diameter device. (d) Cross-sectional schematic of the device architecture, highlighting the layer composition. (e) Cross-sectional STEM image of the via, showing the layer stack including the AlScN ferroelectric layer and HfO_2 insulating layer. (f) High-resolution TEM image of the AlScN/Al interface, illustrating the material quality and interface characteristics. (g) FFT pattern of image shown in panel (e), confirming the crystallographic orientation of the AlScN layer. (h) EDS mapping of the TEM cross-section, displaying the elemental distribution of aluminum, scandium, hafnium, titanium, gold, and oxygen across the device structure.

ferroelectric-metal (MFM) structure without an interfacial insulator. We observed an increase in breakdown field and consistent current density as the device dimensions were reduced. For devices scaled to a diameter of 50 nm, we achieve an ultrahigh breakdown field to coercive field ratio exceeding 2.6, significantly improving reliability and operating range.

Beyond binary switching, we demonstrate stable 3-bit multistate operation in our scaled FeDiodes. This multistate capability is maintained with clear distinction between resistance states and retention exceeding 5×10^5 s with projected stability beyond 10 years. By achieving 40 nm scaling while enhancing critical performance metrics, this work establishes AlScN-based FeDiode memory as a highly promising candidate for next-generation nonvolatile storage with direct integration potential for advanced computing architectures.

A schematic depiction of the Au (120 nm)/Ti (30 nm)/HfO_2 (4 nm)/AlScN (20 nm)/Al(111) (50 nm) MIFM FeDiode with a via structure appears in Figures 1a and 1d. The 20-nm-thick AlScN film is deposited via cosputtering from separate Al and Sc targets onto an in-situ-grown (111)-textured Al layer over a 4-in. sapphire wafer to reduce lattice mismatch. The top surface initially receives another in situ Al layer to prevent oxidation of the ferroelectric. After wet etching, electron-beam lithography (EBL) defines markers, which are patterned by e-beam evaporation of Ti/Au and subsequent liftoff, followed by the deposition of a 100-nm SiO_2 layer, which is then dry etched to form the via. A 4-nm HfO_2 insulating layer is then deposited via atomic layer deposition (ALD) for the MIFM structure, after which the Ti/Au top electrode is evaporated and patterned using liftoff. A detailed fabrication process appears in the Supporting Information (SI) (Figure S1). The fabrication of MFM FeDiodes follows the same procedure, except without the ALD insulator deposition step.

The via structures are characterized using scanning electron microscopy (SEM), as shown in Figure 1c. SEM images confirm successful scaling down to 50 nm. Figure 1e shows a representative cross-sectional scanning transmission electron microscopy (STEM) image of a 4-μm-diameter MIFM device. STEM analysis confirms the uniformity of the insulating and ferroelectric layers, with the measured thicknesses of AlScN and HfO_2 being 20 and 4 nm, respectively. The high-resolution STEM image shown in Figure 1f reveals a sharp and uniform interface without any dead layer between AlScN and Al. A (0002) oriented AlScN has been grown over Al(111). A 2−3 nm oxidized region was also observed between AlScN and HfO_2. This oxidized layer, while potentially beneficial in suppressing leakage through passivation, may also contribute to device-to-device variability and influence long-term endurance and reliability. Figure 1g shows a Fast Fourier Transform (FFT) from the AlScN/Al interface, confirming the crystallographic orientation relationship between AlScN and Al. The AlScN thin film adopts a wurtzite crystal structure, resulting in a measured c/a ratio of 1.56. Figure 1h provides an energy-dispersive X-ray spectroscopy (EDS) analysis of the diode via region, verifying the elemental distribution of respective elements in corresponding thin films.

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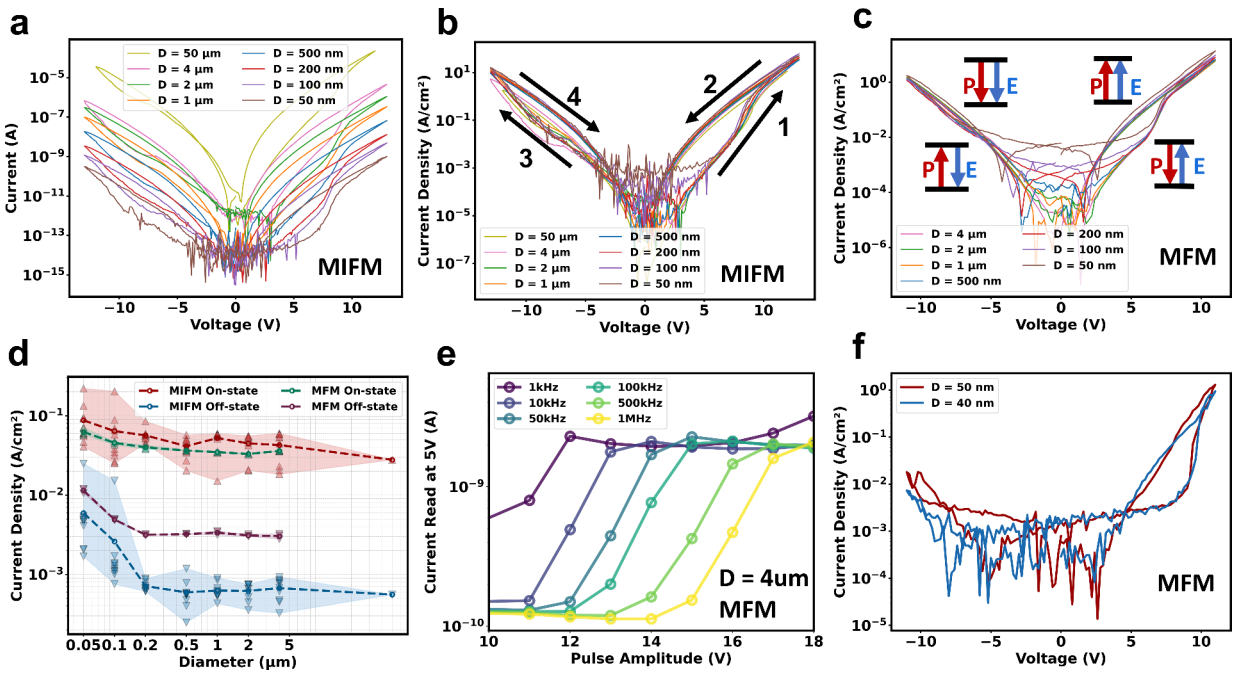


Figure 2. DC electrical characterization of MIFM and MFM structures across different device sizes. (a) $I-V$ characteristics of MIFM structures for various diameters. (b) Current density−voltage characteristics of MIFM and MFM structures. (c) $I-V$ characteristics of MFM structures across different diameters. (d) Current density comparison of ON-state and OFF-state for both MIFM and MFM structures, with statistical variations. (e) AC pulse program with DC resistance readout, demonstrating successful switching up to 1 MHz for MFM devices with $D = 4\,\mu$m. Current was measured at 5 V after each pulse sequence. (f) DC $I-V$ characteristics of a 40-nm device, compared to a 50-nm device from the same chip, showing consistent current density, confirming reliable scaling down to sub-50-nm dimensions.

To systematically evaluate the impact of FeDiode scaling on memory characteristics, we investigate its electrical performance, including switching voltage, leakage behavior, and breakdown reliability. Figure 2 presents the direct-current (DC) electrical measurements of both MIFM and MFM structures across a wide range of device diameters, demonstrating their nonvolatile memory operation and scalability. Quasi-DC (0.01 Hz) current−voltage (I−V) hysteresis measurements are performed with diameters ranging from 50 μm to 50 nm, as shown in Figures 2a−c. The applied maximum voltages of 12 V for MIFM and 11 V for MFM ensure complete ferroelectric switching.

The $I-V$ characteristics of MIFM devices (Figure 2a) show clear switching with a stable ON-OFF ratio maintained for voltages above 5 V. Figure 2b presents the current density as a function of voltage, showing strong consistency across different device sizes at voltages above 5 V. The peak current densities align well, confirming robust fabrication accuracy.

For MFM devices, the current density characteristics (Figure 2c) similarly exhibit consistent scaling trends, with all device sizes demonstrating matching current densities. However, since MFM devices lack an insulating HfO_2 layer, the maximum ON-OFF ratio is reduced, aligning with previous observations.[5] The current characteristics of MFM devices, along with additional DC $I-V$ curves, are provided in the SI (Figure S4).

As shown in Figures 2a−c, the DC $I-V$ characteristics exhibit a clear distinction between the high-resistance state (HRS) and low-resistance state (LRS) before reaching the switching voltage of 7−9 V. As a ferroelectric memory, the FeDiode demonstrates a relatively large memory window compared to HZO-based ferroelectric memory, attributed to AlScN's high coercive field and large remanent polarization.[8] As shown in Figure 2f, the scaled 40 nm diameter device exhibits robust and uniform performance. Considering its device area of 0.00126 μm^2 and operation as a 1 bit memory cell, our FeDiode achieves a potential storage density of 225 Mbit/mm^2, surpassing commercial NOR Flash by more than 20 times.

Figure 2d compares the extracted current density between the ON-state and OFF-state for both MIFM and MFM structures. The MFM devices exhibit high uniformity across all sizes, suggesting that the observed variation in MIFM devices originates from the presence of the HfO_2 insulating layer. Despite this variability, the insulating layer effectively suppresses OFF-state leakage current by reducing tunneling probability. We verified this by modeling the tunneling current with the Poole−Frenkel model in both the ON and OFF states, as shown in SI (Figure S8 and Note 1). Additionally, the insulating layer enhances the on-state current density, leading to an increased ON−OFF ratio.

Observing the ON-state across all sizes, which is less susceptible to measurement noise, reveals a slightly higher current density in smaller devices. This behavior is attributed to local perimeter electric field reinforcement,[9] which becomes more pronounced as the device scales down. The variation in MFM devices remains negligible, indicating that the ferroelectric properties of AlScN are preserved during scaling down to a diameter of 50 nm.

To evaluate high-speed write capability, triangular pulses of varying frequencies up to 1 MHz were applied to program the device, followed by a 5 V DC read to eliminate frequency-induced noise. As shown in Figure 2e, the read current increases progressively with pulse amplitude and saturates once the device is fully programmed.

To directly confirm ferroelectric switching behavior, we perform pulse switching and Positive-Up−Negative-Down

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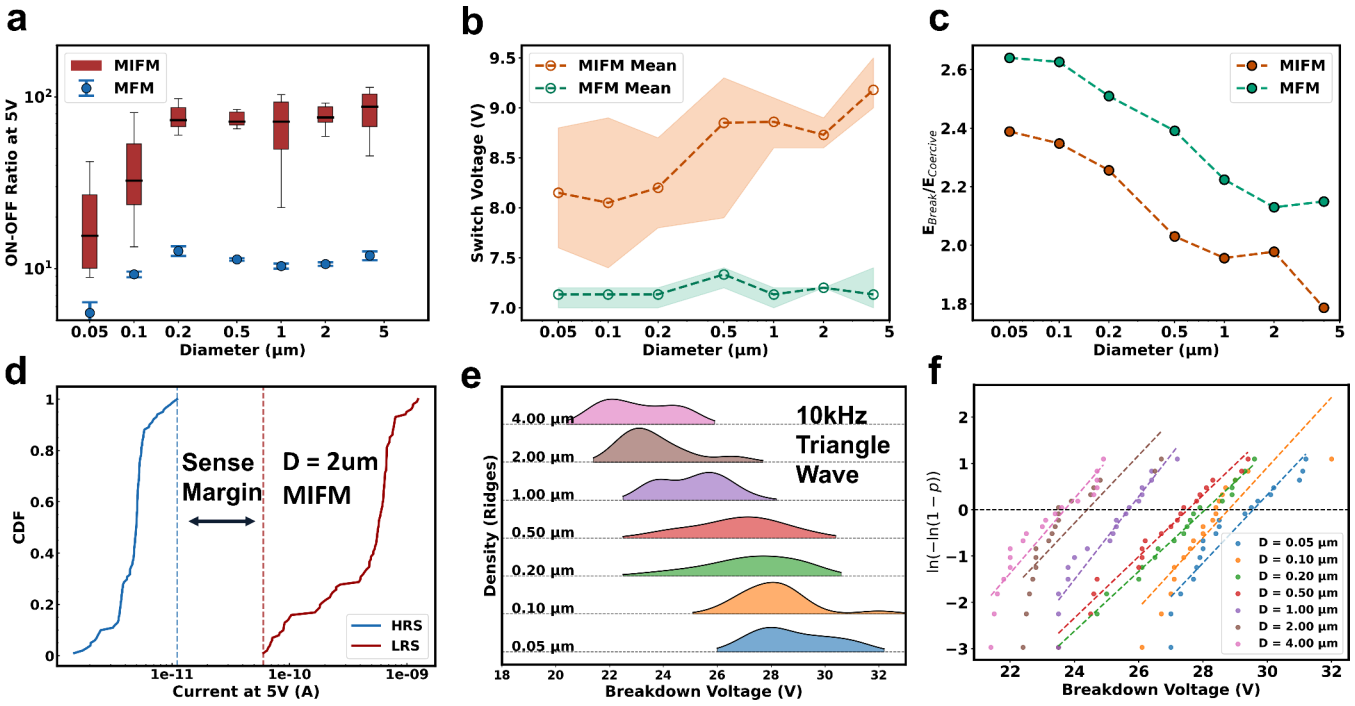


Figure 3. Performance comparison of MIFM and MFM structures in terms of ferroelectric field characteristics and breakdown behavior. (a) ON-OFF ratio at 5 V across various device sizes. (b) DC ferroelectric switching voltage, illustrating size-dependent trends. (c) Ratio of breakdown field to coercive field. (d) Cumulative distribution function (CDF) of 100 MIFM devices with $D = 2\ \mu$m, showing distinct separation between HRS and LRS current at 5 V, confirming consistent switching behavior and a reliable sensing margin. (e) MIFM Breakdown voltage distribution for 20 devices of each size under 10 kHz operation. (f) Weibull plot and corresponding linearization to visualize the statistical reliability of MIFM breakdown voltage across different sizes.

(PUND) polarization measurements. Results clearly indicate ferroelectric polarization reversal, with a saturated 2Pr of approximately 220 μC/cm^2, shown in the SI (Figure S17).

By applying a read voltage of 5 V, the ON-state and OFF-state currents were extracted from 20 DC $I-V$ measurements per size and used to determine the ON-OFF ratio, as shown in Figure 3a. When comparing the performance of MIFM and MFM structures, the addition of the insulating layer enhances state differentiation, particularly in smaller devices. This observation is consistent with previous reports showing that the insertion of an insulator modulates the electrostatic potential barrier and helps suppress direct tunneling.[5]

Figure 3b shows a consistent switching voltage for the MFM structure, corresponding to a DC coercive field of approximately 3.6 MV/cm. The gradient plot used to determine the switching voltage is shown in the SI (Figure S5). The varying via hole diameter for the MIFM device varies the growth rate of the HfO_2 layer, causing a slightly thinner (and more variable) HfO_2 layer for smaller-diameter devices and a reduced switching voltage.

The ratio of the dielectric breakdown field to the coercive field is shown in Figure 3c. As the device dimensions are reduced, the breakdown field increases while the coercive field decreases, leading to a higher breakdown-to-coercive-field ratio. For MFM devices, the measured ratio aligns with reported values at a diameter of 4 μm but exceeds 2.6 when scaled down to 50 nm, demonstrating an improvement over existing reports.[4,10,11] This enhancement suggests that device scaling plays a critical role in improving performance metrics, potentially due to increased electric field effects and improved charge modulation at reduced dimensions.

The measured 100 MIFM devices with a diameter of 2 μm exhibit a clear sense margin and excellent uniformity, as shown in Figure 3d. Additional efforts toward crossbar two-state verification of selector-free operation are presented in the SI (Figure S21), where the yield and performance will be further optimized in future works.

The breakdown voltages of 20 MIFM devices for each size are presented in Figure 3e. A 10 kHz triangular bipolar waveform was applied, with increments of 0.1 V until breakdown occurred. Once a device reaches breakdown, short-circuit behavior and visible metal damage can be observed, as shown in the SI (Figure S7). A key benefit of scaling down is the increase in breakdown field, consistent with previous studies.[11,12]

The Weibull distribution of breakdown field, shown in Figure 3f, provides statistical insight into the reliability of the devices. The results reveal a clear trend where smaller device sizes consistently exhibit higher breakdown voltages. For the 50-nm-diameter devices, the electric breakdown field at 63% probability reaches 12.2 MV/cm.

While two-state (1-bit) switching provides a reliable foundation for nonvolatile memory applications for simple, long-term data storage, the ability to store multiple intermediate states further enhances the storage density and computational potential of FeDiodes. The introduction of an insulating layer in FeDiode enhances its suitability for multistate operation by increasing the ON-OFF ratio and reducing the switching slope of the $I-V$ curves. Unlike conventional memory architectures, FeDiodes offer an efficient approach to processing-in-memory, reducing the need for additional selectors or memristors for complex computations.[6,13]

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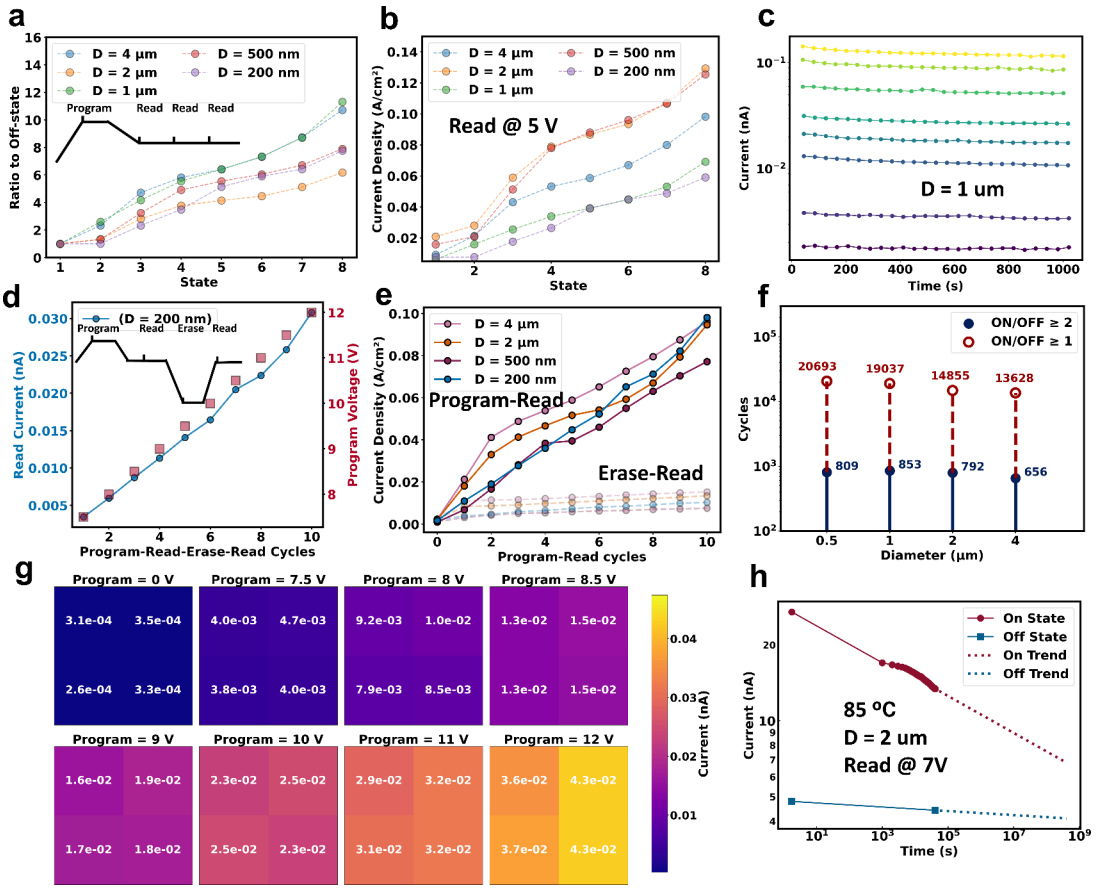


Figure 4. Multistate behavior of MIFM devices across various sizes down to D = 200 nm. (a) Ratio to the OFF-state for each programmed state over multiple program-read cycles, with program voltages ranging from 7.5 to 12 V. (b) Current density of each programmed state, showing distinct multilevel characteristics. (c) Retention behavior of each state over time for a D = 1 μm device, demonstrating data stability. (d) Read current at 5 V for a D = 200 nm device during program−read−erase−read cycles, with erase voltages of −11 V. (e) Current read-after program and erase operations for different device sizes, showing consistency across scaling. (f) AC-programmed, DC-read endurance test. Devices maintaining ON/OFF > 2 are marked in blue and ON/OFF > 1 in red. Results show improved endurance with reduced device diameter. (g) Color map representation of the output current at 5 V for four D = 200 nm devices over program−read−erase cycles, visualizing multistate conductance levels as a function of programming voltage. (h) Two-state retention test at 85 °C, illustrating the stability of the ON-state and OFF-state currents over time.

The multistate behavior of the MIFM device was verified by increasing the program voltage from 7 to 12 V in 0.1 V steps, as shown in the SI (Figure S9). As illustrated in Figures 4a and 4b, 3-bit (8 states) multistate behavior is demonstrated for devices larger than 200 nm in diameter, where both the ratio to the OFF-state and the current density increases progressively from the lowest state to the fully programmed state (state 8). The measurement was conducted using DC program voltages, followed by three consecutive DC read operations at 5 V to ensure accuracy and repeatability. Detailed test procedures are provided in the SI (Figure S10). For sub-200-nm devices, unstable low read signals (see the SI (Figure S11)) were improved using a 7 V read voltage, enabling multistate behavior to 50 nm (see the SI (Figure S12)).

Here, when operating as a multistate device, each programmed state retains its value for over 1000 s at room temperature, as shown in Figure 4c. The measured 1-μm-diameter device exhibits clear and distinguishable differences read continuously at 5 V, further confirming that the multistate behavior arises from partial switching of the ferroelectric AlScN layer.

In addition to the increasing programming voltage with different read current responses, a more rigorous approach was applied to demonstrate the reconfigurable multistate memory behavior of the MIFM FeDiode. As shown in Figure 4d, a program-read-erase-read sequence was implemented on 200 nm devices to assess its performance across multiple cycles. The detailed test setting is shown in the SI (Figure S14). The read current gradually increases in a linear manner, indicating a linear conductance variation across different states. This linearity is crucial for low-power neural network computing, enabling efficient analog computation in artificial intelligence hardware.[14]

As shown in Figure 4e, devices >200 nm show clear, size-independent multistate behavior, confirming ferroelectric AlScN scaling. For smaller devices, the 5 V readout is noise-limited; 7 V results are given in the SI (Figure S15). To evaluate device-to-device variation, we test four separate 200-nm devices, with the results shown in Figure 4g. While variation is clearly present across devices, the programmed states remain reasonably distinguishable. To account for this variability conservatively, the storage density projection is based on reliable two-bit operation, ensuring practical scalability and robustness.

Since it is difficult to accurately measure 2Pr in the small devices, the endurance shown in Figure 4f is estimated using

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Table 1. Benchmarking between High-Density Memory Technology

memory technology	SRAM	DRAM	ferroelectric (HZO) DRAM	NOR flash	MRAM	PCRAM	RRAM	FeDiode	
cell structure	6T	1T-1C	1T-1C	1Floating Gate	1T-1R	1T-1R	1T-1R	1FeDiode (built-in selector)	
cell size	$>100F^2$	$>6F^2$	$>4F^2$	$>10F^2$	$>6F^2$	$>4F^2$	$>4F^2$	$4F^2$	
process node (nm)	$F \approx 20$	$F \approx 30$	$F \approx 24$	$F \approx 100$	$F \approx 100$	$F \approx 100$	$F \approx 90$	$F = 40$	$F = 20$
storage density (Gb/mm^2)	0.038	0.3	0.225 (per layer)	0.01	0.0125	0.05	0.02 (per layer)	0.225 (1 bit)	0.625 (1 bit)
nonvolatility	no	no	yes	yes	yes	yes	yes	yes	
ON/OFF ratio	N/A	N/A	N/A	10^4–10^5	~2.5	10^2	10	10^2	
write voltage	<1 V	<1 V	1.5 V	>10 V	<0.5 V	<3 V	<3 V	<10 V	
retention	N/A	milliseconds to seconds	>10 years	>10 years	>10 years	>10 years	>10 years	>10 years	
reference	18	19	20	21	22	23	24	this work	

an AC programming and DC readout approach. The corresponding data, detailed in the SI (Figure S18), confirm stable device operation over repeated switching cycles. Notably, we observe a scaling benefit in the form of increased endurance at smaller dimensions, which we attribute to the enhanced breakdown-to-coercive field ratio enabled by dimensional scaling.

We evaluated the stability of two resistance states at elevated temperature (85 °C) for over 4×10^4 seconds, ensuring long-term data retention. Additionally, the FeDiodes show retention performance comparable to FeFETs[15] with a simpler fabrication process and improved CMOS compatibility.

Finally, we compare and benchmark our FeDiode technology to various other volatile and nonvolatile memory technologies in Table 1. We demonstrate the scalability of our FeDiode device, achieving a storage density of 225 Mb/mm^2 at the 40 nm node under a $4F^2$-limited crossbar array architecture, with the potential to reach 625 Mb/mm^2 for the 20 nm node, while maintaining stable single-bit operation. The storage density is calculated in the SI (Note 2). Although selector-free operation is already well-demonstrated for FeDiode architectures,[16] we additionally include a selector-free crossbar feasibility analysis based on our experimentally measured rectification behavior in the SI (Note 3) to further support the integration potential.

Although the switching voltage is higher than in some emerging memory technologies, it can be further reduced through optimization of ferroelectric thickness, doping concentration, and contact engineering. These improvements can enhance performance while maintaining the high integration density and selector-free operation of the AlScN FeDiode. Its key novelty lies in device scaling with invariant properties, unlike RRAM and PCRAM, which exhibit increased variability and thermal challenges respectively as dimensions shrink.[17]

While the fabrication process remains relatively complex and the ON/OFF ratio can be improved, these limitations are addressable through further material and structural refinement. This study demonstrates meaningful progress toward scalable integration, with ongoing efforts focused on lowering switching voltage, improving multistate stability at nanoscale dimensions, and evaluating endurance and variability at the array level.

To conclude, we have demonstrated the first successful scaling of AlScN-based ferroelectric diode memory to 40-nm dimensions, a storage density of 225 $Mbit/mm^2$ for a device area of 0.00126 μm^2 at 1-bit operation, establishing a new paradigm for high-density, nonvolatile storage. Our systematic investigation reveals a remarkable size-dependent enhancement in device performance where switching voltage decreases and breakdown field increases as dimensions shrink, resulting in an ultrahigh breakdown-to-coercive field ratio of 2.6 at 50 nm. The MIFM architecture, incorporating a thin HfO_2 insulating layer, enables stable 3-bit storage with distinct resistance states, demonstrating the potential for high-density multibit storage and compute-in-memory functionality in selector-free crossbar arrays. Our FeDiodes maintain exceptional retention characteristics, with demonstrated stability over 4×10^4 seconds at 85 °C and projected retention beyond 10 years. This work highlights the potential of BEOL-compatible AlScN for scalable ferroelectric memory, offering multistate operation and integration readiness for stand-alone, embedded, and compute-in-memory applications.

■ ASSOCIATED CONTENT

Data Availability Statement

All data are available in the paper and Supporting Information. Additional data are available from the corresponding author upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.nanolett.5c02961.

Materials and methods; ferroelectric characterization including PUND and frequency-dependent hysteresis; AC program−DC read retention and endurance measurements; temperature-dependent tests; device-to-device variability and CDF analysis; statistical switching voltage extraction; real crossbar demonstration; and selector-free calculation based on measured data (PDF)

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Notes

The authors declare no competing financial interest.

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